```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
```

```
gap = px.data.gapminder()
gap_2007 = gap[gap["year"] == 2007]   # only the latest year

gap.head()
```

	country	continent	year	lifeExp	pop	gdpPercap	iso_alpha	iso_num
0	Afghanistan	Asia	1952	28.801	8425333	779.445314	AFG	4
1	Afghanistan	Asia	1957	30.332	9240934	820.853030	AFG	4
2	Afghanistan	Asia	1962	31.997	10267083	853.100710	AFG	4
3	Afghanistan	Asia	1967	34.020	11537966	836.197138	AFG	4
4	Afghanistan	Asia	1972	36.088	13079460	739.981106	AFG	4

Part A: Bar Charts

Task 1: Ranked bar chart

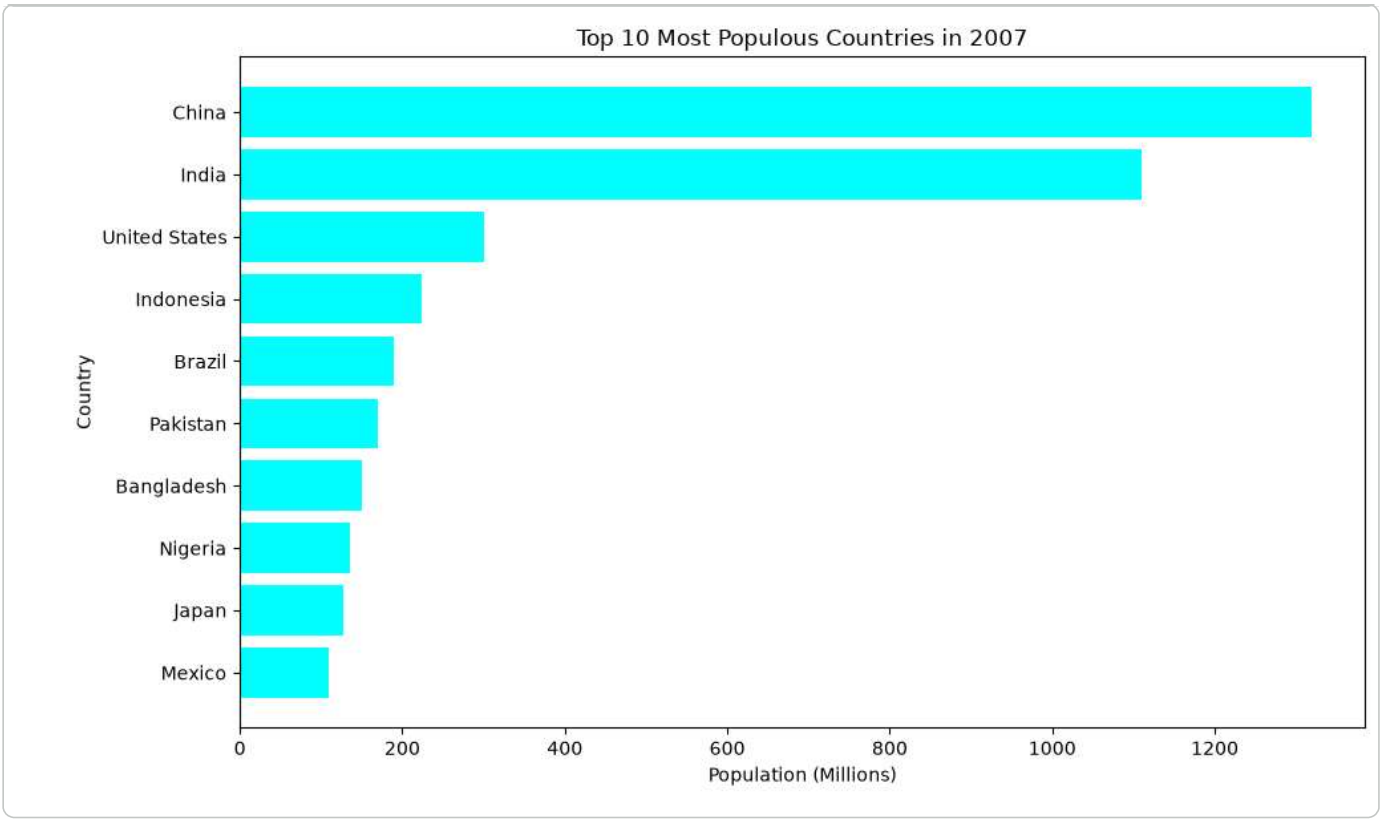
```
import matplotlib.pyplot as plt
import pandas as pd

# 1. Filter the 10 countries with the largest population in 2007
top10_pop = gap_2007.nlargest(10, "pop").copy()

# Convert population to millions
top10_pop["pop_m"] = top10_pop["pop"] / 1_000_000

# Sort in ascending order so the largest bar appears at the top of plt.barh()
top10_pop_sorted = top10_pop.sort_values(by="pop", ascending=True)

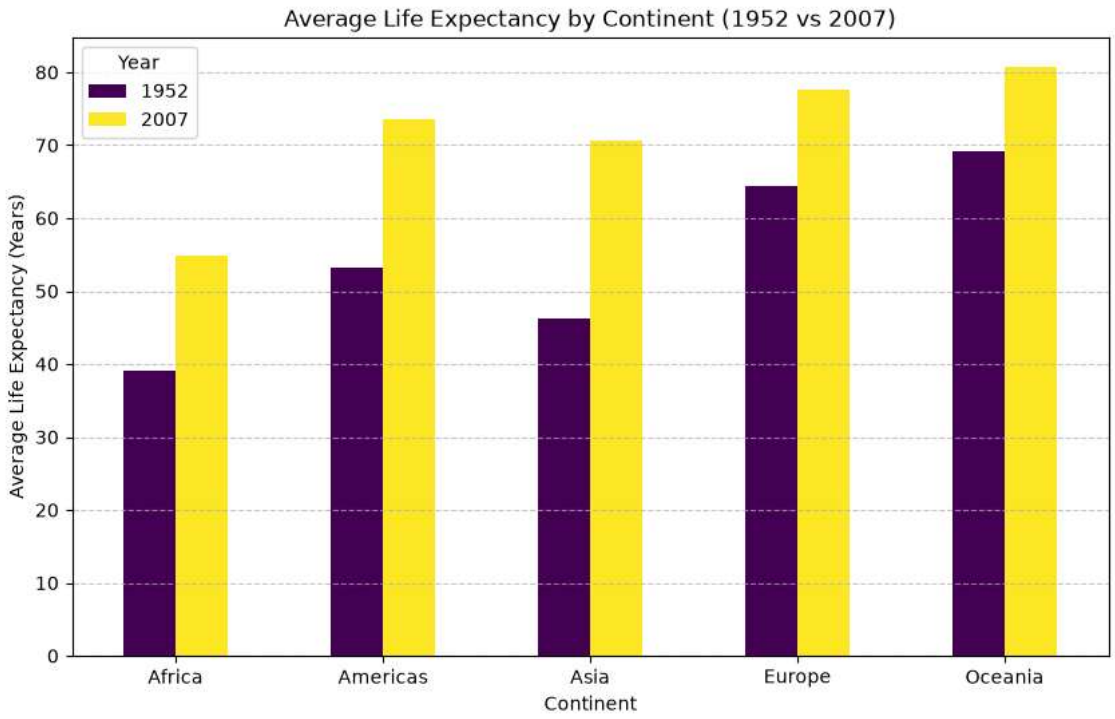
# 2. Draw horizontal bar chart
plt.figure(figsize=(10, 6))
plt.barh(top10_pop_sorted["country"], top10_pop_sorted["pop_m"], color="aqua")
plt.xlabel("Population (Millions)")
plt.ylabel("Country")
plt.title("Top 10 Most Populous Countries in 2007")
plt.tight_layout()
plt.show()
```



Task 2: Grouped bar chart

```
import matplotlib.pyplot as plt
import pandas as pd
import plotly.express as px
import seaborn as sns
gap = px.data.gapminder()
g = (
    gap[gap["year"].isin([1952, 2007])]
    .groupby(["continent", "year"])["lifeExp"]
    .mean()
    .unstack()
)

# 3. Draw the grouped bar chart
g.plot(kind="bar", figsize=(10, 6), colormap="viridis")
plt.xlabel("Continent")
plt.ylabel("Average Life Expectancy (Years)")
plt.title("Average Life Expectancy by Continent (1952 vs 2007)")
plt.xticks(rotation=0)
plt.legend(title="Year")
plt.grid(axis="y", linestyle="--", alpha=0.7)
plt.show()
```

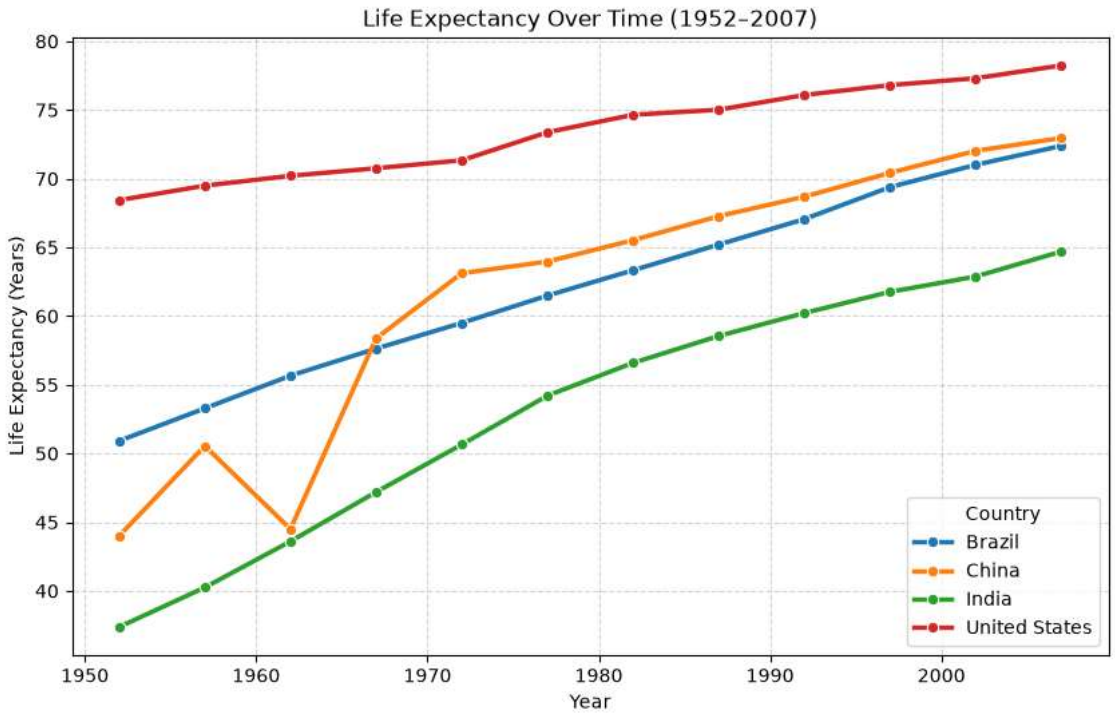


Part B: Line Chart

Task 3: Change over time

```
# Filter data
countries = ["India", "China", "Brazil", "United States"]
subset_data = gap[gap["country"].isin(countries)]

plt.figure(figsize=(10, 6))
sns.lineplot(
    data=subset_data,
    x="year",
    y="lifeExp",
    hue="country",
    marker="o",
    linewidth=2.5,
)
plt.xlabel("Year")
plt.ylabel("Life Expectancy (Years)")
plt.title("Life Expectancy Over Time (1952-2007)")
plt.legend(title="Country")
plt.grid(True, linestyle="--", alpha=0.5)
plt.show()
```



Part C: Scatter Plots

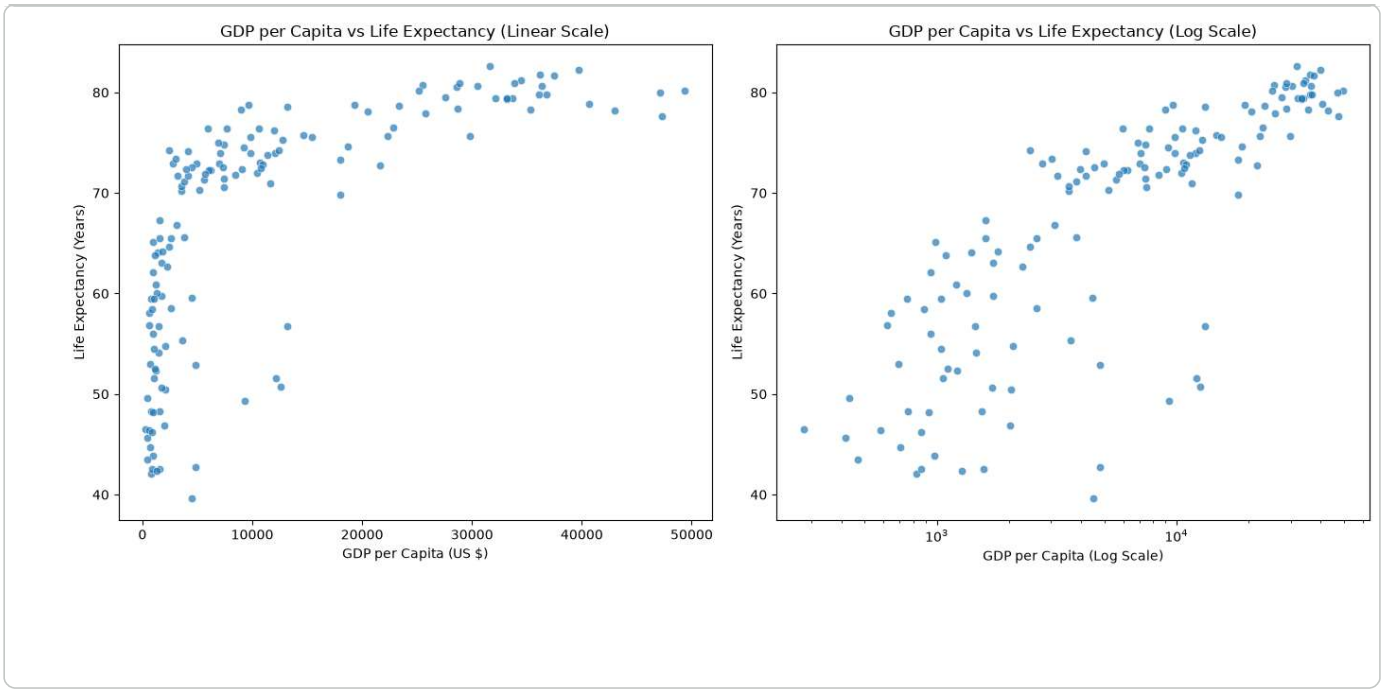
Task 4: Linear scale versus log scale

```
fig, axes = plt.subplots(1, 2, figsize=(14, 6))

# Left chart: Linear scale
sns.scatterplot(
    data=gap_2007, x="gdpPercap", y="lifeExp", ax=axes[0], alpha=0.7
)
axes[0].set_title("GDP per Capita vs Life Expectancy (Linear Scale)")
axes[0].set_xlabel("GDP per Capita (US $)")
axes[0].set_ylabel("Life Expectancy (Years)")

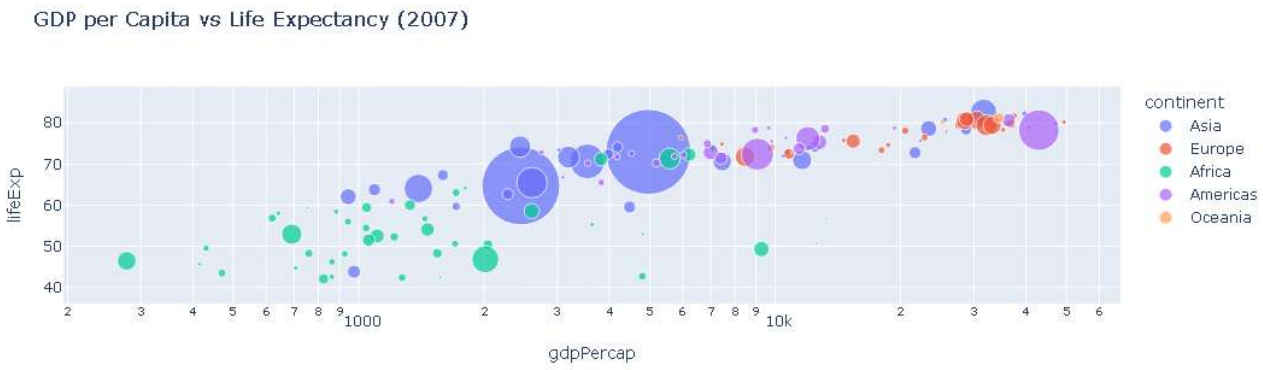
# Right chart: Log scale
sns.scatterplot(
    data=gap_2007, x="gdpPercap", y="lifeExp", ax=axes[1], alpha=0.7
)
axes[1].set_xscale("log")
axes[1].set_title("GDP per Capita vs Life Expectancy (Log Scale)")
axes[1].set_xlabel("GDP per Capita (Log Scale)")
axes[1].set_ylabel("Life Expectancy (Years)")

plt.tight_layout()
plt.show()
```



Task 5: Bubble chart

```
fig = px.scatter(
    gap_2007,
    x="gdpPercap",
    y="lifeExp",
    size="pop",
    color="continent",
    hover_name="country",
    log_x=True,
    size_max=50,
    title="GDP per Capita vs Life Expectancy (2007)",
)
fig.show()
```



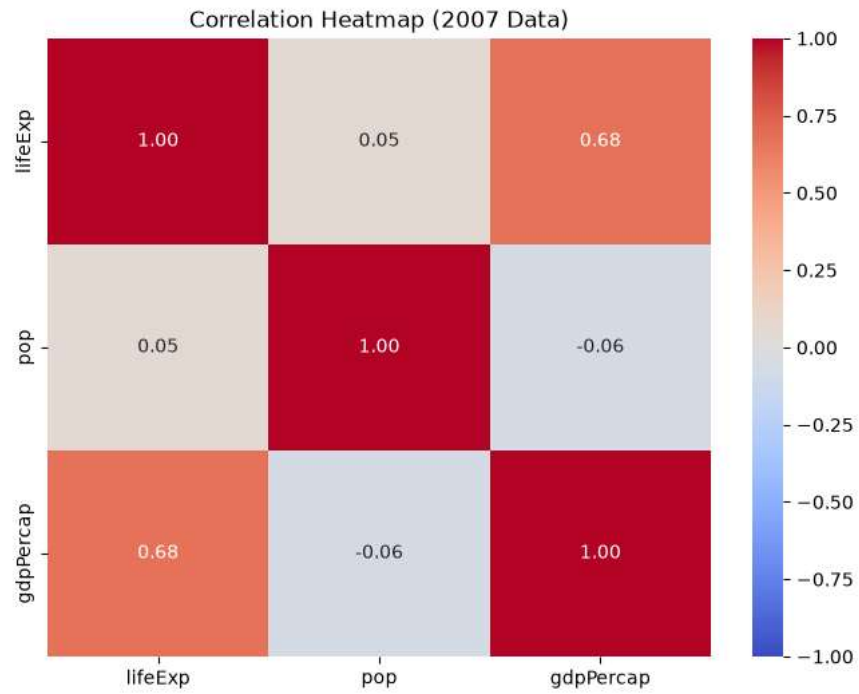
Part D: Heatmaps

Task 6: Correlation heatmap

```
# Calculate correlation matrix
corr_matrix = gap_2007[["lifeExp", "pop", "gdpPercap"]].corr()

plt.figure(figsize=(8, 6))
sns.heatmap(
```

```
    corr_matrix, annot=True, cmap="coolwarm", vmin=-1, vmax=1, fmt=".2f"
)
plt.title("Correlation Heatmap (2007 Data)")
plt.show()
```



Task 7: Pattern heatmap

```
pivot = gap.pivot_table(
    index="continent", columns="year", values="lifeExp", aggfunc="mean"
)

plt.figure(figsize=(10, 6))
sns.heatmap(pivot, annot=True, fmt=".0f", cmap="YlGnBu")
plt.title("Average Life Expectancy by Continent and Year")
plt.xlabel("Year")
plt.ylabel("Continent")
plt.show()
```

Part E: Choosing and Improving Charts

Task 8: Choose the right chart

Situation	Best chart	Reason
Compare the number of students in five branches	Bar chart	Best for comparing categorical discrete values side-by-side.
Show how a city's temperature changed month by month in a year	Line chart	Ideal for tracking continuous trends and changes sequentially over time.
Check whether more study hours go with higher marks	Scatter plot	Perfect for displaying the relationship or correlation between two numerical variables.
Show how the marks of 100 students are spread out	Histogram	Designed specifically to show the distribution and frequency spread of a single continuous variable.
Show attendance of 5 subjects over 6 weeks in one view	Heatmap	Excellent for visualizing two-dimensional matrix patterns and intensity across categories and time.

Task 9: Fix a poor chart

